A Paleomagnetic Study of Some Tertiary and Quaternary Rocks of Yamaguchi Prefecture, Japan

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Abstract. Samples of Quaternary basaltic rocks and Tertiary sediments of Yamaguchi Prefecture, West Japan, have been collected and their natural remanent magnetizations and other magnetic properties measured. The mean directions of the natural remanent magnetization of the Quaternary basalts is nearly parallel to the present geomagnetic field, and that of the Tertiary sediments of Miocene age also agrees with the present geomagnetic field, but the Paleogene sediments show very large deviations in the mean direction of their remanent magnetization from the present geomagnetic field. Some reversals of the remanent magnetization in situ were found among the samples of early Quaternary age. Virtual paleogeomagnetic dipole positions have been estimated from these data for Tertiary and Quaternary times in Yamaguchi Prefecture, West Japan.

Introduction. Some magnetic properties of samples of Yamaguchi basalts and sediments have been measured previously [*Asami and Domen,* 1954, 1955; *Domen,* 1960]. The natural remanent magnetizations of most of these rocks agreed well with the present geomagnetic field, but a few of them showed so-called reverse natural remanent magnetization.

This paper presents a summary of these data on Yamaguchi basalts and on some Tertiary sediments. Virtual paleogeomagnetic pole positions for Tertiary and Quaternary times in Yamaguchi Prefecture, West Japan, have been estimated. The reversals of the natural remanent magnetization found among the basalts erupted during the early Quaternary are also discussed. In the near future, more details of paleomagnetic studies on early and pre-Tertiary times in this area will be published elsewhere.

Sampling and experimental techniques. About three hundred samples were collected from the basaltic bodies of the northern part of Yamaguchi Prefecture, West Japan. Thirty samples came from Susa Tertiary layers to the northeast of the above mentioned basalts, and thirteen came from the Tertiary deposit located on the coast of the Seto Inland Sea at the famous Ube coal mines. From each oriented sample collected, several 2- to 3-cm cubes were cut; sometimes the roughly spherical hand specimens were used as test specimens.

The magnetometer used was an ordinary astatic magnetometer with a 30-cm stem supporting two oppositely directed, well-aged N. K. S. steel magnets, 0.5 cm in diameter and 1 cm in length. This astatic magnet system was suspended by two different sizes of thin phosphor-bronze strips connected end to end. The lower strip has a torque constant of about 0.5 dyne cm/radian and is about 10 cm long. This ribbon supports the magnet system at its lower end and is fixed at the upper end to the other strip, which has a torque constant of about 0.05 dyne cm/radian and is about 100 cm long. By clamping the ribbons at the point where they are joined, the sensitivity of the magnetometer can easily be changed, say from 10^{-6} to 10^{-4} emu per scale division.

The specimen is mounted on a sample holder placed either just below or in front of the lower magnet and is rotated through 30° steps in a horizontal plane by remote control in order to obtain a series of measurement from which its remanent magnetic moment can be computed.

Natural remanent magnetizations of Tertiary and Quaternary rocks. Most Yamaguchi basalts appear to have been erupted during and after the Pleistocene, but the absolute chronology of these lava flows is not known and even their relative ages are difficult to determine. Petrological and chemical investigations of the Ceno-

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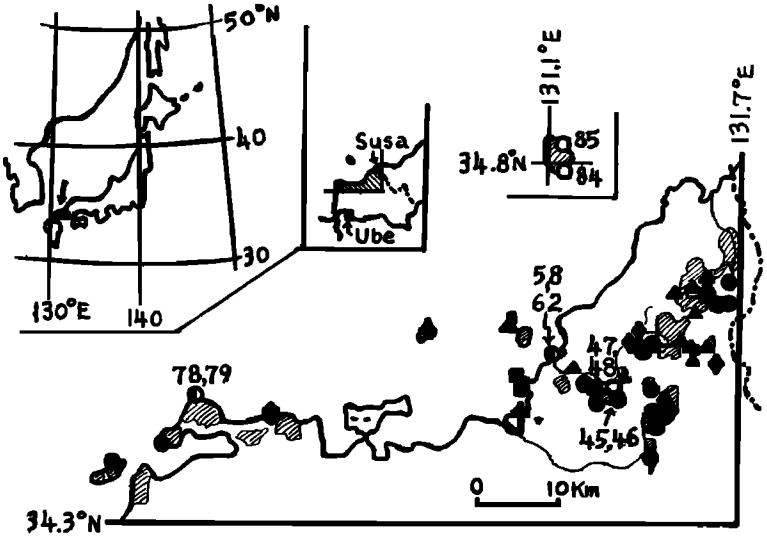


Fig. 1. Locations and sampling sites of Yamaguchi basalts. Square, group 1; triangle, group 2; diamond, group 3; circle, group 4; hexagon, group 5. Open symbols indicate sites at which reverse NRM's were found. Solid symbols indicate normal NRM's. Numerals are given only for those sites at which reverse NRM's were found. Inset shows the localities of Ube and Susa Tertiary deposits.

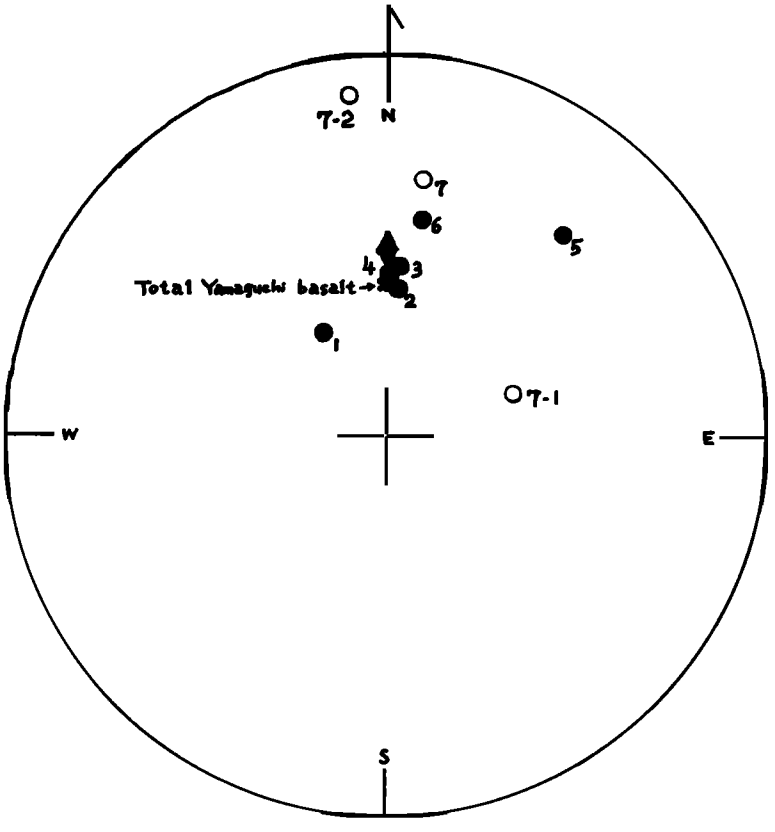


Fig. 2. Mean direction of NRM of Yamaguchi Tertiary sediments and Quaternary basalts. Numerals indicate group numbers. Solid symbols indicate downward dips and open symbols indicate upward dips. Diamond, present geomagnetic field.

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zoic rocks of Yamaguchi Prefecture have recently been carried out [*Kurasawa and Takahashi,* 1960], and on this basis the present author has assumed that Cenozoic volcanic activity in this district commenced with a tholeiitic rock series, followed, roughly, by an alkali and then a calc-alkali rock series.

The Yamaguchi basalts have therefore been classified into five petrological and geological groups as follows:

Group 1, Calc-alkali rocks; Holocene
Group 2, Calc-alkali rocks; Pleistocene (Upper)
Group 3, Alkali rocks; Pleistocene (Middle)
Group 4, Alkali rocks; Pleistocene (Lower)
Group 5, Tholeiitics rocks; Plio-Pleistocene

Figure 1 shows the sampling sites classified into these five groups. The mean directions of the natural remanent magnetizations of groups 2, 3, and 4 fall close to each other and to the present geomagnetic field (Figure 2). Groups 1 and 5 deviated in direction from the above groups.

In Figure 3, the specific intensity of the natural remanent magnetization J_{rn} of groups 1 to 5 are plotted on a log-log scale against the specific susceptibility in the present geomagnetic horizontal field (0.31 oersted). This figure shows a linear correlation between log J_{rn} and log χ (the slope of the line is about 1.5). This agrees with the general relation previously found between J_{rn} and J_{iH} (induced magnetization in the present geomagnetic field) for all the Yamaguchi basalts. The average ratio of intensity of the natural remanent magnetizations and the susceptibility, J_{rn}/χ, of each group is greater than unity. This suggests that most Yamaguchi basalts have rather stable remanent magnetizations [*Kazmi,* 1961].

Concerning the sediments, the Susa Tertiary layers were formed in the Miocene, and the Ube district Tertiary deposition was completed during the Paleogene [*Geological Survey of Japan,* 1956]. The data for the natural remanent magnetizations of samples of these two sediments have also been listed in groups 6 and 7 of Table 1 and plotted in Figure 2. Group 7 has been divided into two sub-groups, 7–1 and 7–2. The sampling locations represented by these groups differ by 180 meters in depth. The mean direction of the natural remanent magnetization of the samples in group 7 has a very large semi-

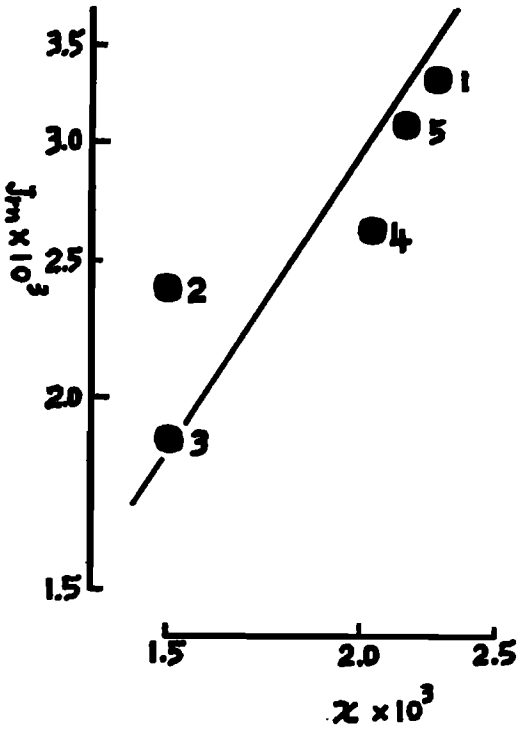


Fig. 3. Correlation of intensity of NRM and susceptibility for Yamaguchi basalts. Numerals are group numbers. The equation of the line is $J_{rn} = 31.6\ \chi^{1.5}$.

angle of the 95% circle of confidence [*Fisher,* 1953]. The Ube Tertiary deposits include several coal beds; samples belonging to group 7–1 from the mountainside are white and rather coarse, but those of group 7–2 from below sea level are reddish sandstones. The mean dip angle of the natural remanent magnetization of group 7–2 is much smaller than that of group 7–1. This suggests that the remanent magnetization of group 7–2 might include a secondary magnetization caused by pressure or chemical remanence [*Domen,* 1957*a,* 1962; *Kawai et al.,* 1959].

The virtual paleogeomagnetic pole positions for the Tertiary and Quaternary of Yamaguchi Prefecture. The paleogeomagnetic virtual pole positions of these Quaternary basalts and Tertiary sediments are listed in Table 1 and plotted on a Schmidt net in Figure 4. The total mean paleogeomagnetic pole position of all Yamaguchi basalts is also shown [*Domen,* 1960; *Irving,* 1961].

The virtual pole positions for groups 1 and 5

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TABLE 1. NRM's of Tertiary and Quaternary Rocks and Virtual Paleogeomagnetic Pole Positions of Yamaguchi Prefecture

Group	Declination (E)	Inclination (D)	α_{95}	J_{rn}, emu/g	χ, emu/g	J_{rn}/J_{sE}	J_{rn}/χ	Virt. Pole Pos.		Number of Samples
1. Calc-alkali basalt, Holocene	−32°	64°	9°	3.3×10^{-3}	2.3×10^{-3}	4.3	1.4	76°E	63°N	31
2. Calc-alkali basalt, Pleistocene (Upper)	4°	58°	6°	2.4	1.5	4.8	1.6	163°E	85°N	140
3. Alkali basalt, Pleistocene (Middle)	4°	53°	24°	1.9	1.5	5.0	1.3	178°E	87°N	19
4. Alkali basalt, Pleistocene (Lower)	1°	55°	4°	2.6	2.1	4.1	1.2	131°E	88°N	209
5. Tholeiitic basalt, Plio-Pleistocene	41°	30°	17°	3.1	2.2	4.6	1.4	123°W	49°N	17
6. Sediment, Miocene	9°	42°	39°	3.6×10^{-6}				89°W	78°N	31
7. Sediment, Paleogene	8°	−32°	69°	2.2×10^{-6}				124°E	38°S	13
7–1	72°	−62°		2.7×10^{-6}				88°E	12°N	5
7–2	−6°	−11°	54°	1.8×10^{-6}				140°E	50°S	8

deviated from the present pole and also from those of groups 2, 3, and 4. The pole positions for groups 2, 3, and 4 are close to each other. It seems likely that these virtual pole positions are representative of Quaternary time for Yamaguchi Prefecture. Group 6, which is Miocene, shows rather good agreement with other data obtained elsewhere in Japan (N. Kawai, 1963, private communication). The virtual pole position of group 7 (Paleogene) deviates from the virtual poles obtained by other authors [*Cox and Doell*, 1960].

Reversals of natural remanent magnetization. Reversals of the natural remanent magnetization have so far been found in this district only in group 4 (lower Pleistocene). Some of the magnetic properties of these reversals are listed in Table 2 with those of the normal polarizations obtained from the same sites. There is also a linear correlation between log J_{rn} and log χ, resembling that for all the Yamaguchi basalts.

The ratios J_{rn}/χ and stability tests by means of ac demagnetization and of heating suggest that the magnetization has remained unaltered since it was first acquired.

The virtual paleogeomagnetic pole positions calculated are listed in Table 2. Both the normal and reversed positions for sites 59 to 62, 78, 79, 84, and 85, are closely grouped, but those for sites 45 to 48 are not. This concentration of the coexisting normal and reversed natural remanent magnetization, i.e. of the virtual paleogeomagnetic pole positions obtained in the small areas and at about the same geological age, suggests that both the normal and reversed magnetization of these specimens are closely related. On the other hand, the samples from sites 45 to 48 were much weathered and their J_{rn}/χ ratios are

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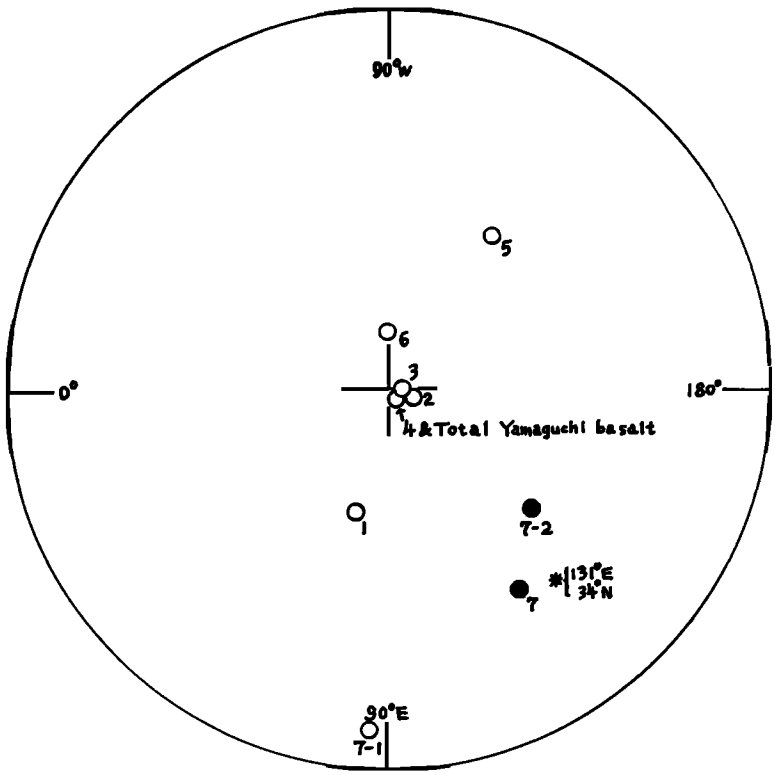


Fig. 4. Virtual paleogeomagnetic pole positions for Tertiary and Quaternary in Yamaguchi Prefecture. Numerals are group numbers. Open circles indicate that the pole position is in northern hemisphere. Solid circles indicate southern hemisphere. Asterisk, sampling site.

TABLE 2. Magnetic Data of Normal and Reverse NRM of Yamaguchi Basalts

Site No.	Declination (E)	Inclination (D)	α_{95}	J_{rn}, emu/g	χ, emu/g	J_{rn}/J_{iE}	J_{rn}/χ	Virt. Pole Pos.		Number of Samples
					Reversed Group					
45	115°	−61°	29°	0.5×10^{-3}	1.5×10^{-3}	1.1	0.33	68°E	39°N*	8
47	140°	−77°	15°	0.8	2.9	1.1	0.28	105°E	52°N*	7
48	−147°	−33°	31°	1.4	3.2	1.7	0.44	118°W	56°N*	3
58	175°	−47°	5°	3.0	2.2	4.5	1.4	14°W	82°N*	2
79	−167°	−55°	7°	1.2	1.1	3.2	1.1	145°W	79°N*	11
84	−173°	−63°	9°	0.6	0.2	11.2	3.0	159°E	79°N*	3
85	−173°	−65°	1°	0.9	0.1	22.	9.0	152°E	77°N*	2
					Normal Group					
46	78°	39°	57°	0.4	1.4	0.9	0.28	115°W	22°N	2
59	−19°	49°	84°	7.7	4.1	6.1	1.9	32°E	74°N	2
60	−3°	32°	9°	8.5	4.3	6.4	2.0	39°W	72°N	5
61	10°	49°	9°	3.0	2.2	4.5	1.4	112°W	80°N	7
62	−3°	28°	5°	5.0	3.1	5.3	1.6	41°W	70°N	4
78	28°	52°	22°	11.3	7.3	4.7	1.6	143°W	68°N	9

* South pole.

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TABLE 3. Paleomagnetic Data for Sites 78 and 79 of Yamaguchi Basalts

Group	Declination (E)	Inclination (D)	α_{95}	J_{rn}, emu/g	χ, emu/g	J_{rn}/J_{iE}	J_{rn}/χ	Virt. Pole Pos.		(1)	Number of Samples
N^2	25°	52°	8°	28.6×10^{-3}	14.8×10^{-3}	9.2	1.9	141°W	69°N*	31%	3
N^3	23°	51°	66°	4.9	6.6	2.3	0.74	134°W	70°N	50	3
N^4	38°	52°	69°	0.4	0.5	2.7	0.80	144°W	58°N	83	3
R^3	−167°	−55°	14°	1.8	1.7	3.5	1.1	143°W	79°N*	14	5
R^4	−167°	−55°	10°	0.5	0.5	3.0	1.0	143°W	79°N*	10	6

(1) Diminution of NRM by ac demagnetization at 200 oe peak.
* South pole.

about one-third. This requires more precise investigations. Reversals at site 79 have been previously reported [*Asami and Domen,* 1955] and thoroughly investigated [*Asami,* 1956]. The present author has also paid much attention to this reversal, and thermomagnetic analysis of these rocks and statistical details have been reported previously [*Domen,* 1956, 1960].

Sites 78 and 79 are in one lava flow; the samples from these sites are classified into five groups which are symbolized by N^2, N^3, N^4, R^3, and R^4. This classification is based on the intensity of the natural remanent magnetization and the direction of polarization of the specimens; i.e., R^3 shows that the intensity of such a group is of the order of 10^{-3} emu/g and the

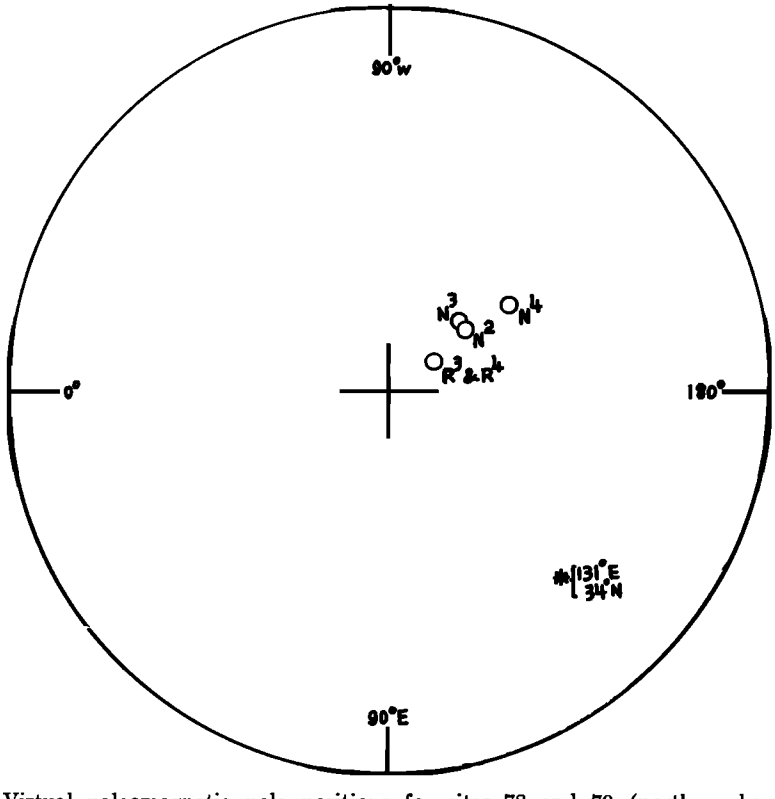


Fig. 5. Virtual paleomagnetic pole positions for sites 78 and 79 (northern hemisphere). Asterisk, sampling site.

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TABLE 4. Summary of Stability of NRM of Sites 78 and 79

	Demagnetization (ac)	J_{rn}/χ	K ratio (J_{rn}/J_s)	Exsolution	Dispersion
N^2	Fairly Stable	Large	Large	Well developed	Small
N^3	Unstable	Small	Medium	Under developed	Large
N^4	Unstable	Small	Small	Under developed	Large
R^3	Very Stable	Medium	Medium	Fairly developed	Small
R^4	Very Stable	Medium	Medium	Developed	Small

direction of the magnetization is reversed. According to this classification, the natural remanent magnetizations and the virtual paleogeomagnetic pole positions and other magnetic properties of these groups are given in Table 3 and in Figure 5. There is a linear correlation between log J_{rn} and log χ, of which the slope is about 1.4. The summary of the stability of each group is given in Table 4. As shown in this table, for the alternating magnetic field the natural remanent magnetization of the reversed groups is more stable than that of the normal groups. The J_{rn}/χ ratios for N^3 and N^4 are rather small, which means that the natural remanent magnetizations of these groups might have been altered since they were first acquired. J_{rn} of N^2 might be stable, or the samples may be young, so that the K ratio [$= J_{rn}/J_s$, *Domen*, 1957*b*; *Kawai et al.*, 1956*a*] for this group is large. In the normal groups the magnitude of the semi-angle of the circle of confidence increases with decrease in the intensity of their natural remanent magnetizations. In reversed groups the directional vectors have converged. Note that the direction of the natural remanent magnetization of reversed groups is nearly antiparallel to those of normal ones. By means of an X-ray analyzer, it has been shown that the contained ferromagnetic minerals of these rocks are magnetites and Ti-poor titanomagnetites. The result of thermomagnetic analysis shows that N^3 and N^4 have unstable phases with an intermediate Curie point, whereas R^3, R^4, and N^2 have stable phases with both lower and higher Curie points (Figure 6), so that it might be possible to apply Kawai's self-reversal theory of the remanent magnetization in the ferromagnetic minerals with the solid phase transformation [*Kawai et al.*, 1954*a*, *b*; 1956*a*, *b*; *Kawai*, 1955, 1956] to this coexistence of the normal and reverse of the natural remanent magnetization.

If these reversals of the NRM's of the Yamaguchi basalts were produced by a reversed geomagnetic field, this would conflict with world-wide data which indicate that the latest geomagnetic field reversal occurred in Plio-Pleistocene time. The reversal groups of Yamaguchi basalts are now thought to be Pleistocene, but the author hopes that radioactive dating techniques will be applied to these specimens in the future to prove or disprove this age.

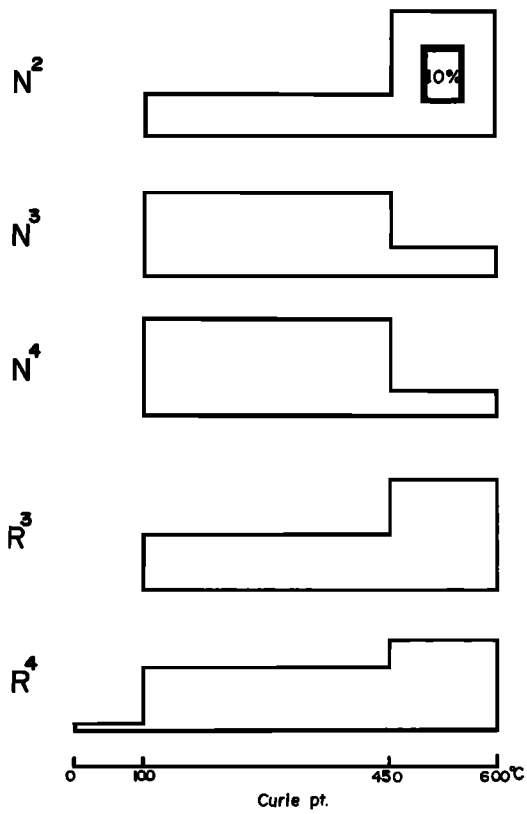


Fig. 6. Phase distribution diagrams.

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